ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CE ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Define Stress, Strain, Young's Modulus, Bulk Modulus, Shear Modulus. Derive relation between E, G, K. [14 Marks]

[OR]

Q.1 (Alt) Calculate stresses in compound bars and thermal stresses in constrained members. [14 Marks]

--- UNIT II ---

Q.2 Draw Shear Force and Bending Moment Diagrams for Cantilever and Simply Supported Beams with UDL and UVL. [14 Marks]

[OR]

Q.2 (Alt) Derive Pure Bending Equation (M/I = sigma/y = E/R). Calculate Section Modulus for I-section. [14 Marks]

--- UNIT III ---

Q.3 Derive Torsion Equation (T/J = tau/r = G theta / L) for circular solid and hollow shafts. [14 Marks]

[OR]

Q.3 (Alt) Design helical springs and calculate deflection, stiffness, and maximum shear stress. [14 Marks]

--- UNIT IV ---

Q.4 Explain Principal Stresses and Principal Planes. Construct Mohr's Circle of Stress for 2D states. [14 Marks]

[OR]

Q.4 (Alt) Explain Theories of Failure: Maximum Principal Stress, Maximum Shear Stress (Tresca), Von Mises. [14 Marks]

--- UNIT V ---

Q.5 Derive Euler's Critical Buckling Load for columns with various end boundary conditions. [14 Marks]

[OR]

Q.5 (Alt) Explain Rankine-Gordon formula for columns and thin cylinders circumferential/longitudinal stresses. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****